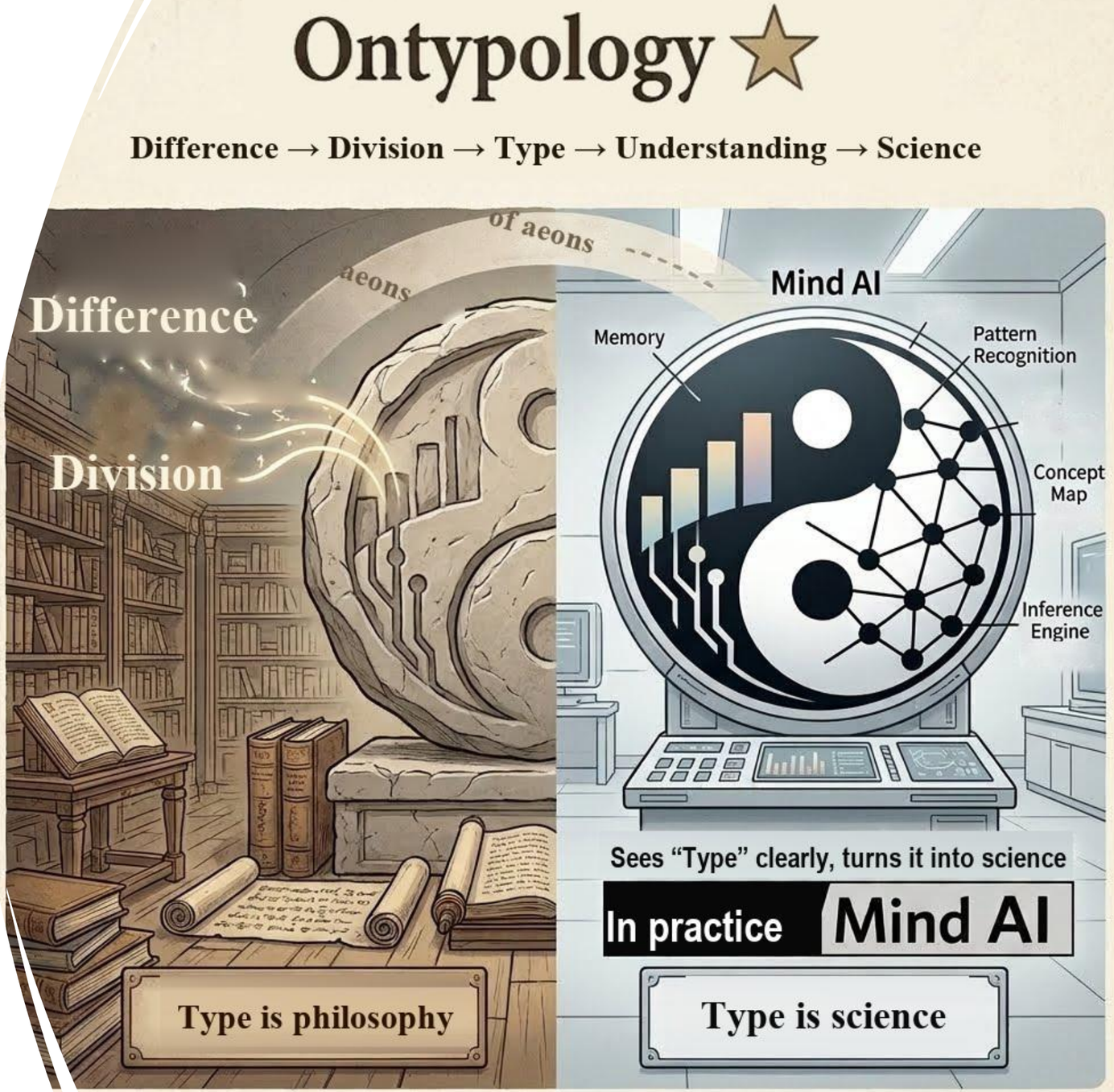
Ontypology

- From difference to division, from division to type, from type to understanding - the brain has walked this path for aeons.
- A century ago, logic was philosophy; once seen clearly, it became an established science.
- Today we explore "Type" (a brain concept) as philosophy; Ontypology makes "Type" clear, turns it into an established science, and puts it into practice as Mind AI.

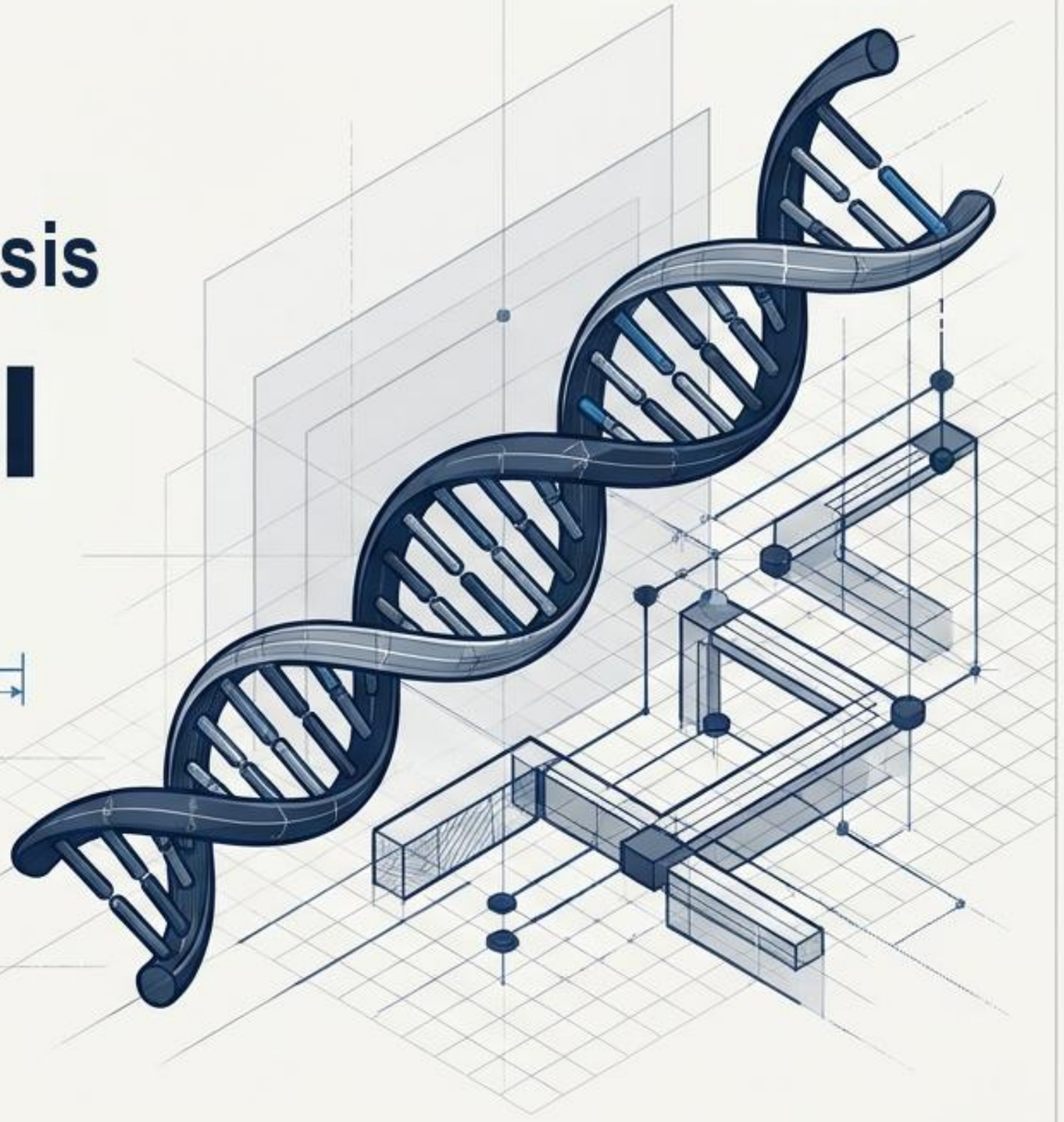


Genome-wide association analysis

GWAS × Mind AI

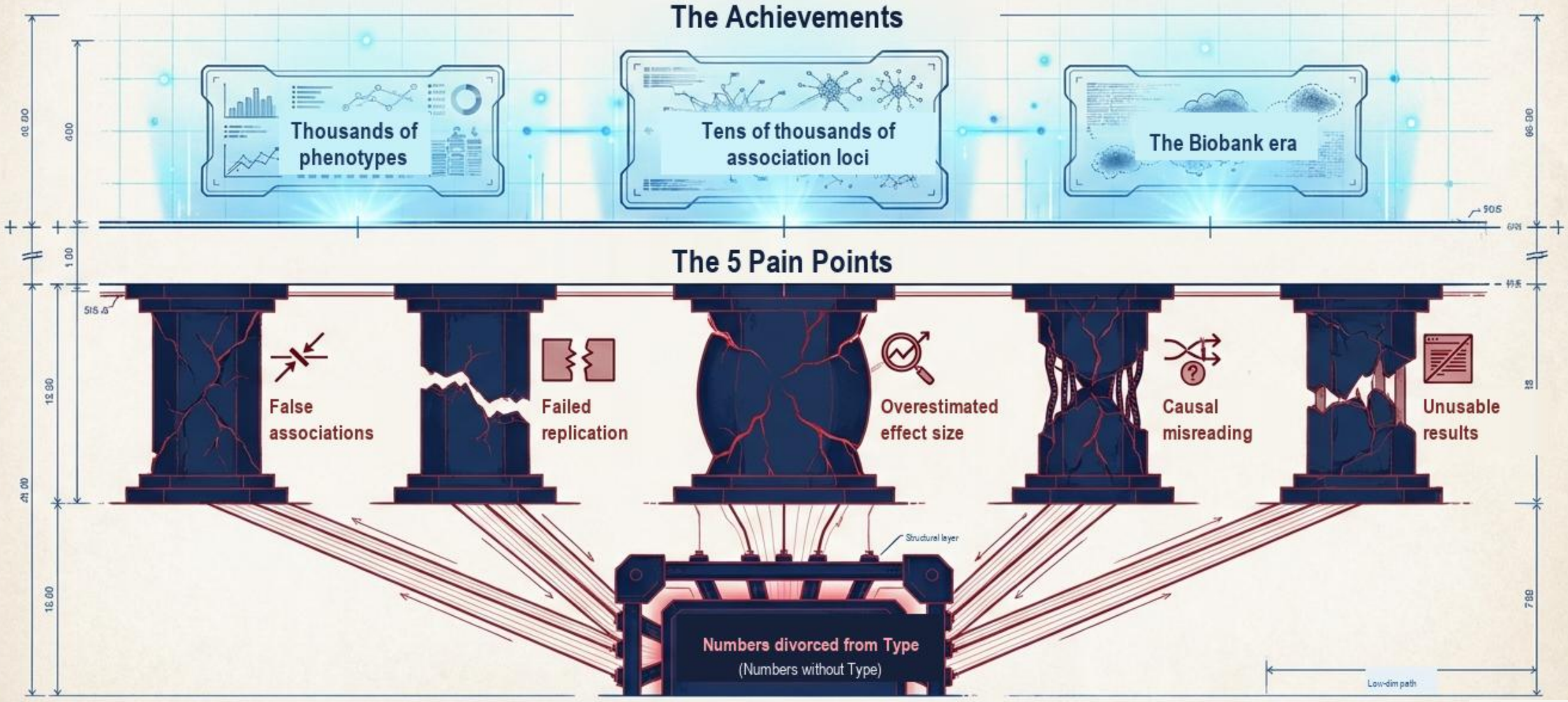
Accurate usage, architecture, and a complete content plan

Types first, then numbers



The fatal wound behind a powerful discovery engine

We computed the probabilities right — and got the levels wrong.



The systemic collapse of GWAS and its five pain points

PAIN POINT 1

False associations

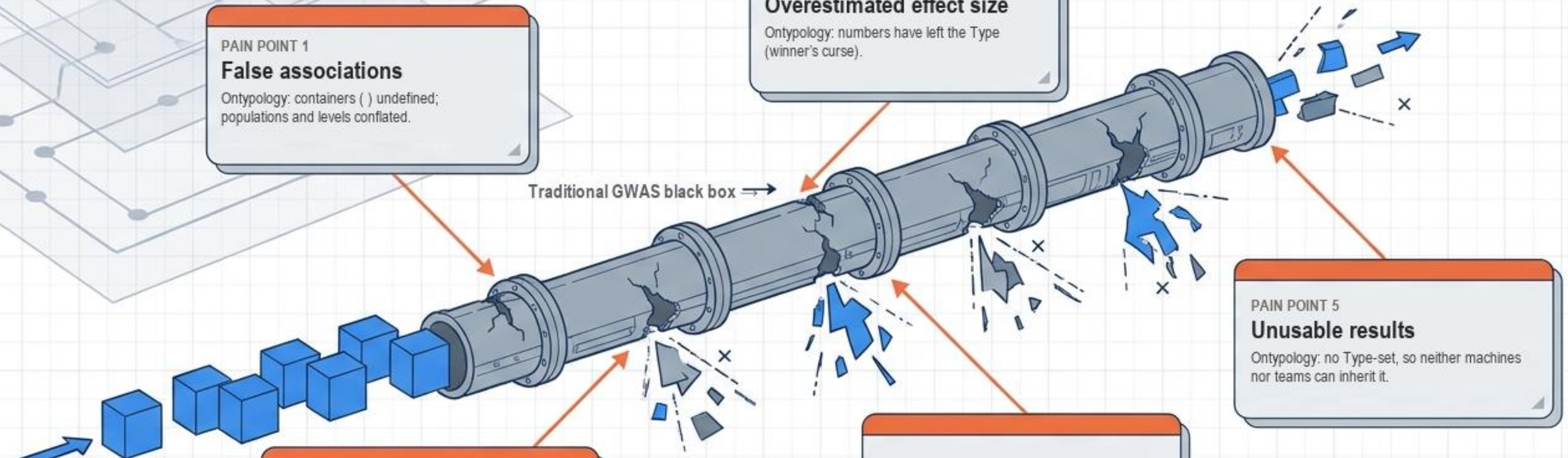
Ontypology: containers ( ) undefined; populations and levels conflated.

PAIN POINT 3

Overestimated effect size

Ontypology: numbers have left the Type (winner's curse).

Traditional GWAS black box ⟹



PAIN POINT 5

Unusable results

Ontypology: no Type-set, so neither machines nor teams can inherit it.

PAIN POINT 2

Failed replication

Ontypology: the [ ] rules do not carry through; criteria incomplete.

PAIN POINT 4

Causal misreading

Ontypology: association ≠ causation; the Time Type is unfinished.

The problem is not statistics; it is the lack of a descriptive language that makes levels explicit.

MIND AI

Cognitive shift: Type first, numbers second

A number is a sub-Type inside a Type, a secondary comparison. Without the Type as base, a p-value cannot say which container computed it — an invalid number.

Old thinking: floating numbers

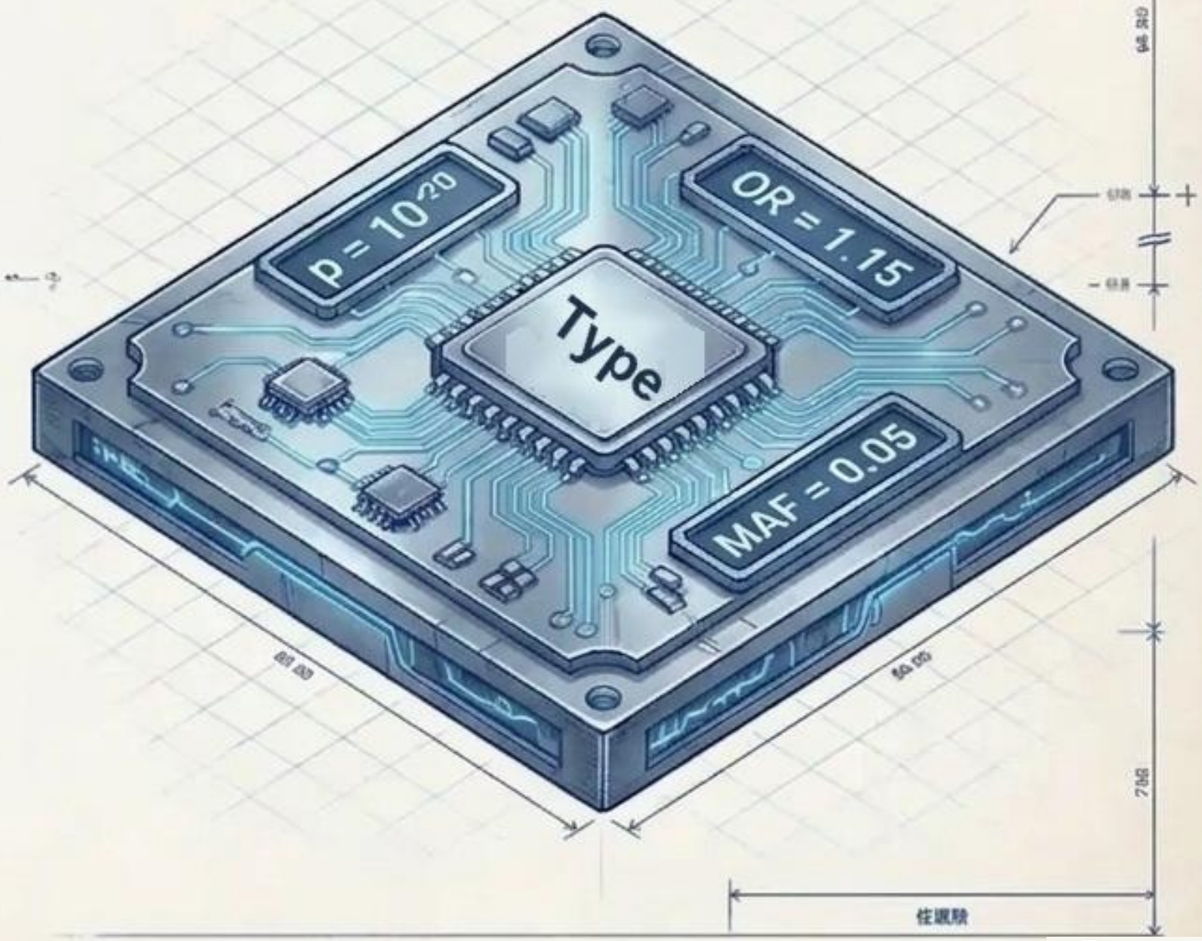
Prone to the winner's curse

$p = 10^{-20}$

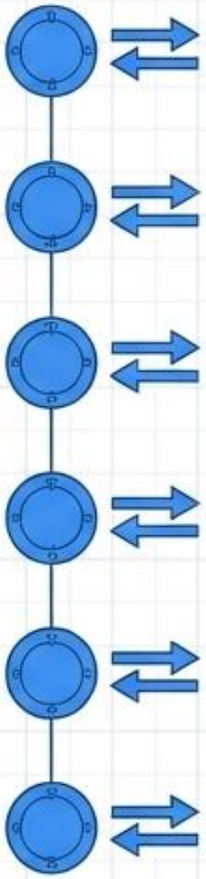
OR = 1.15

MAF = 0.05

New thinking: grounded type



GWAS ↔ Ontypology: architecture mapping table

Study unit (cohort individual)		**Unit Type: { individual, (genotype), [phenotype] }**
Phenotype		**X-axis attribute [ ]: dimension × scale**
Variant / SNP		**Y-axis composition { }: genome parts and three remove levels**
Cohort / population		**Z-axis container ( ): container C = (id(C), S(C))**
Analysis pipeline		**Time Type: function chain (Type in, Type out)**
Significance threshold (e.g., 5×10^{-8})		**Three-level test: abstraction criterion ∩(T)**

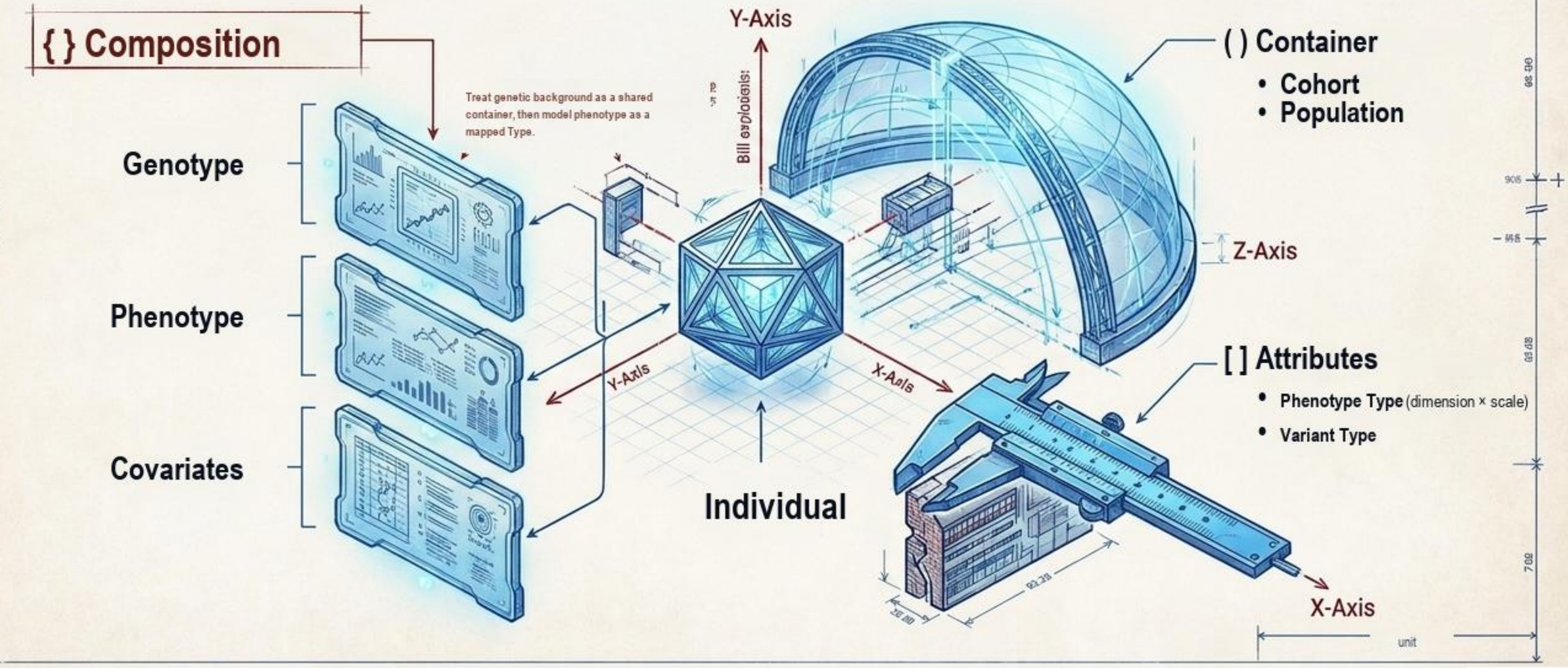
Core thesis: a number is a sub-Type inside a Type. Every p-value and effect size must attach to the Types above.

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Spatial Type deconstructed: the minimal complete set for GWAS results

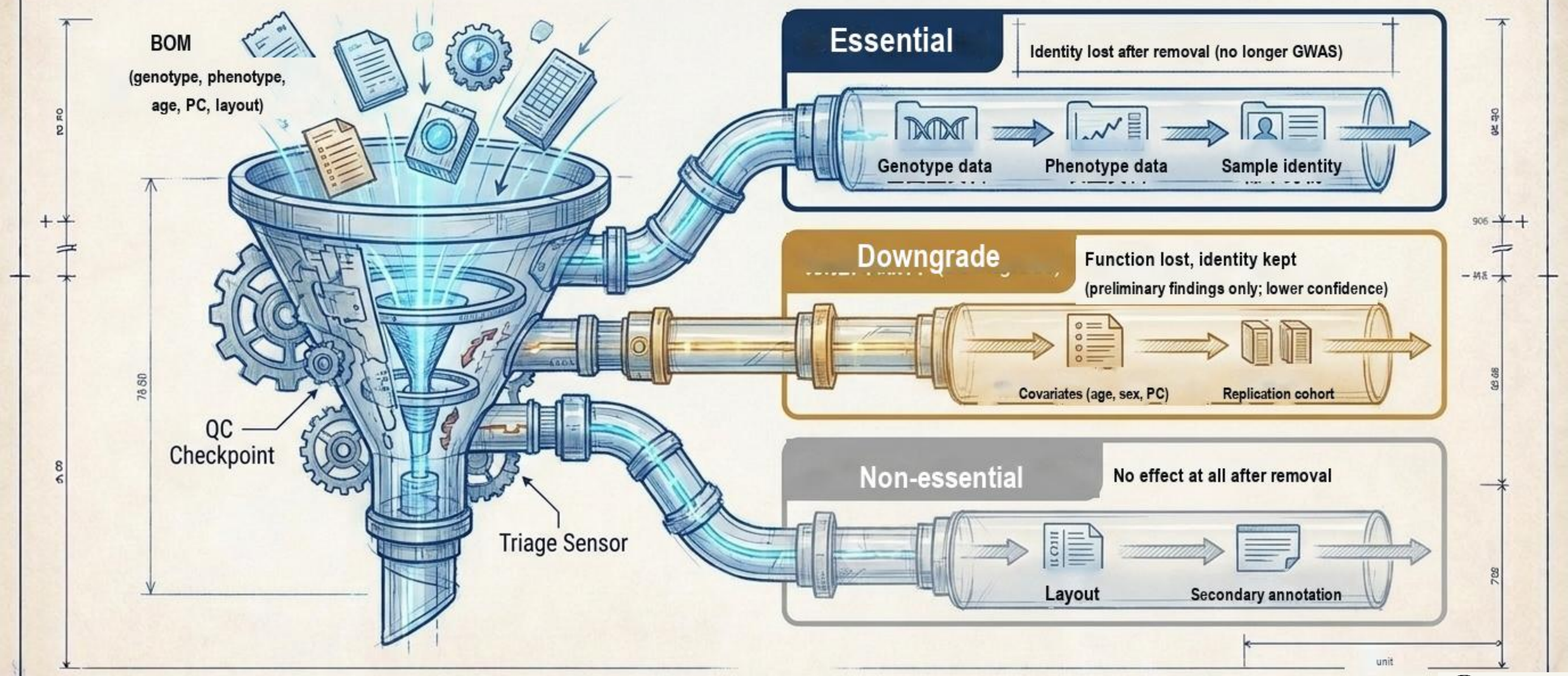
Type-set notation: { individual, (genotype), [phenotype] } — define the entry spec first, so data arrives as rows.

Study-object Unit Type



Deep dive into the Y-axis: QC is where the Remove criterion is applied

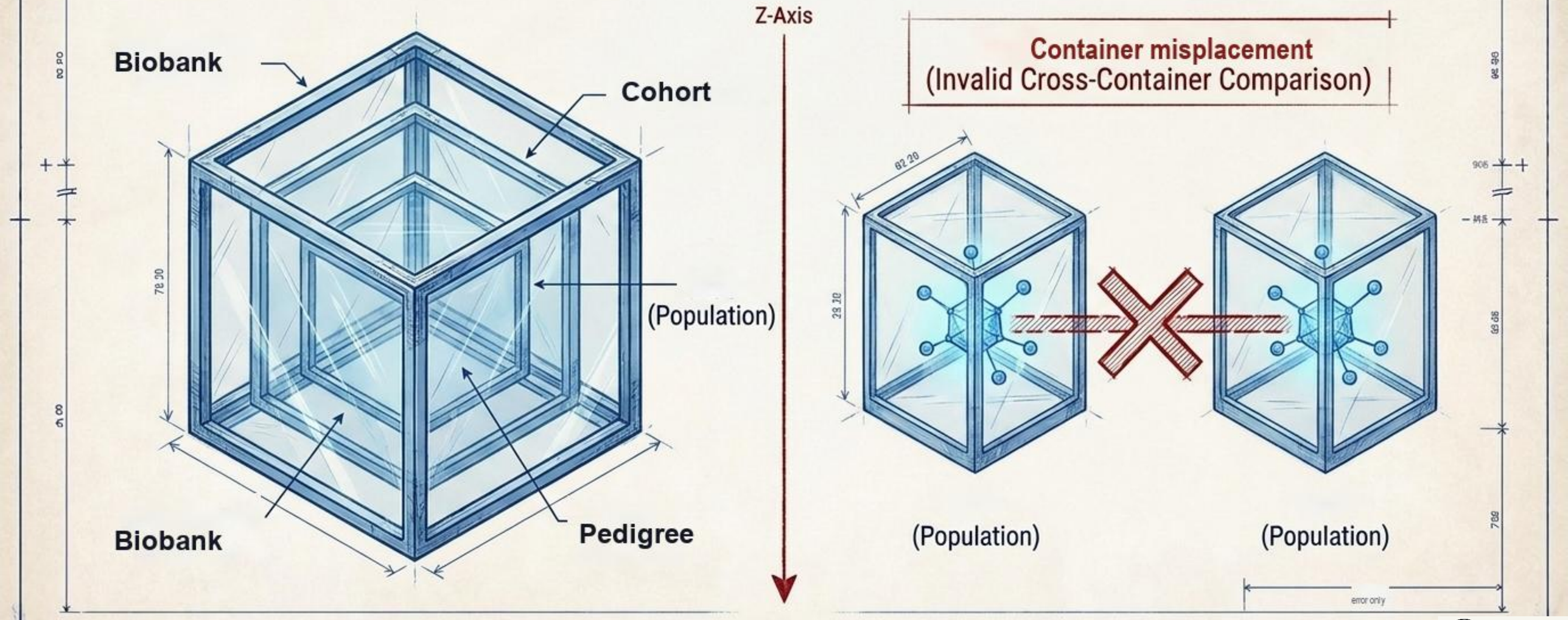
Every removal rule answers one question: is this sample a damaged essential part, or noise from a non-essential one?



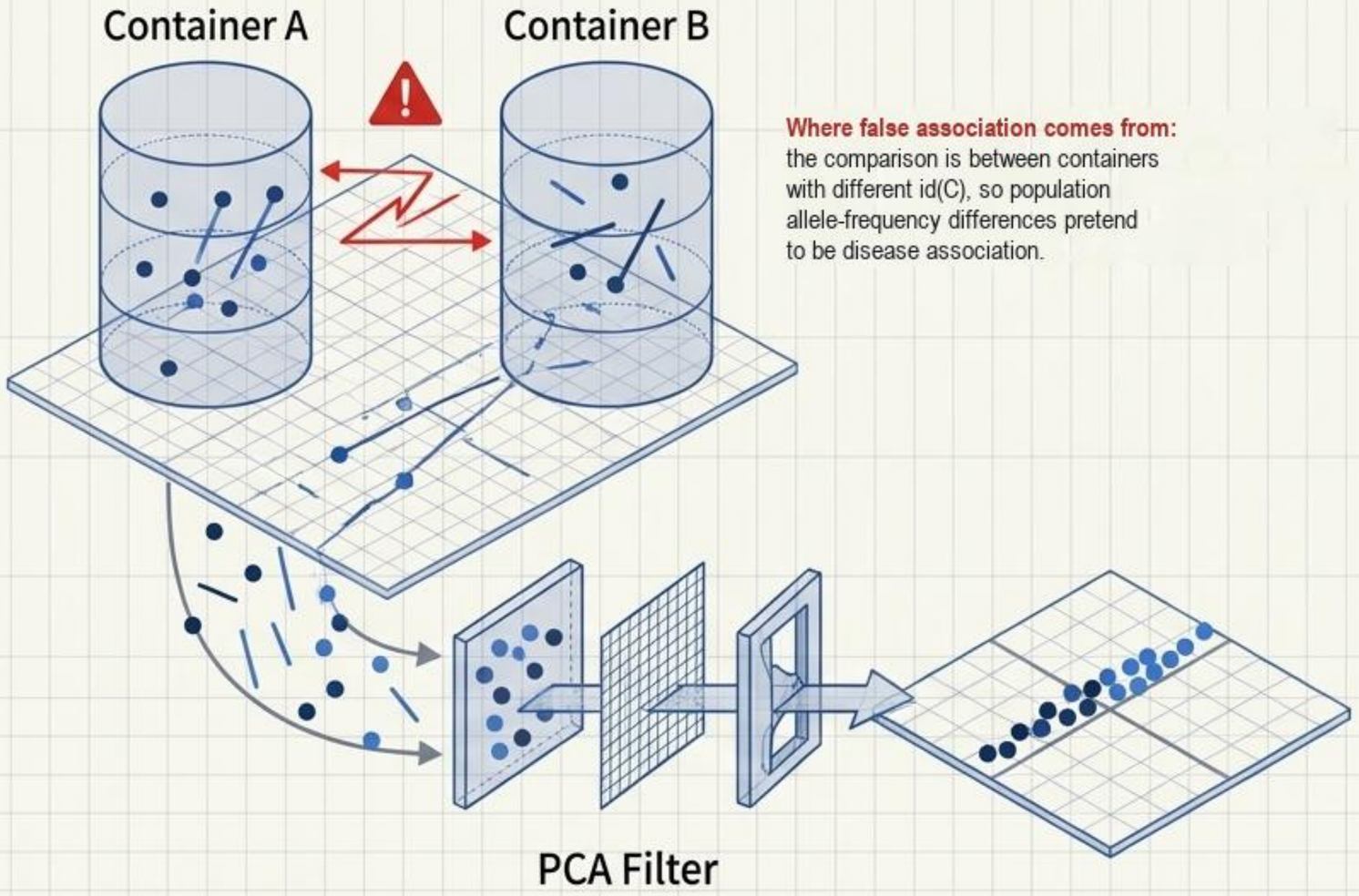
Z-axis dive: stratification and false association are really ‘container misplacement’

You think you are comparing associations inside a container, but you are comparing two different containers.

Solution: C = (id(C), S(C)). PCA and mixed models essentially subtract the ‘container effect’ from the comparison.



Deeper into the Z-axis: container effects and population stratification



Where false association comes from:
the comparison is between containers with different id(C), so population allele-frequency differences pretend to be disease association.

Statistical genetics filter

Container formula:
$C = (id(C), S(C))$
- identity and internal space.

HWE & MAF:
Quality and frequency baseline inside the container.

LD (linkage disequilibrium):
The spatial structure inside the container.

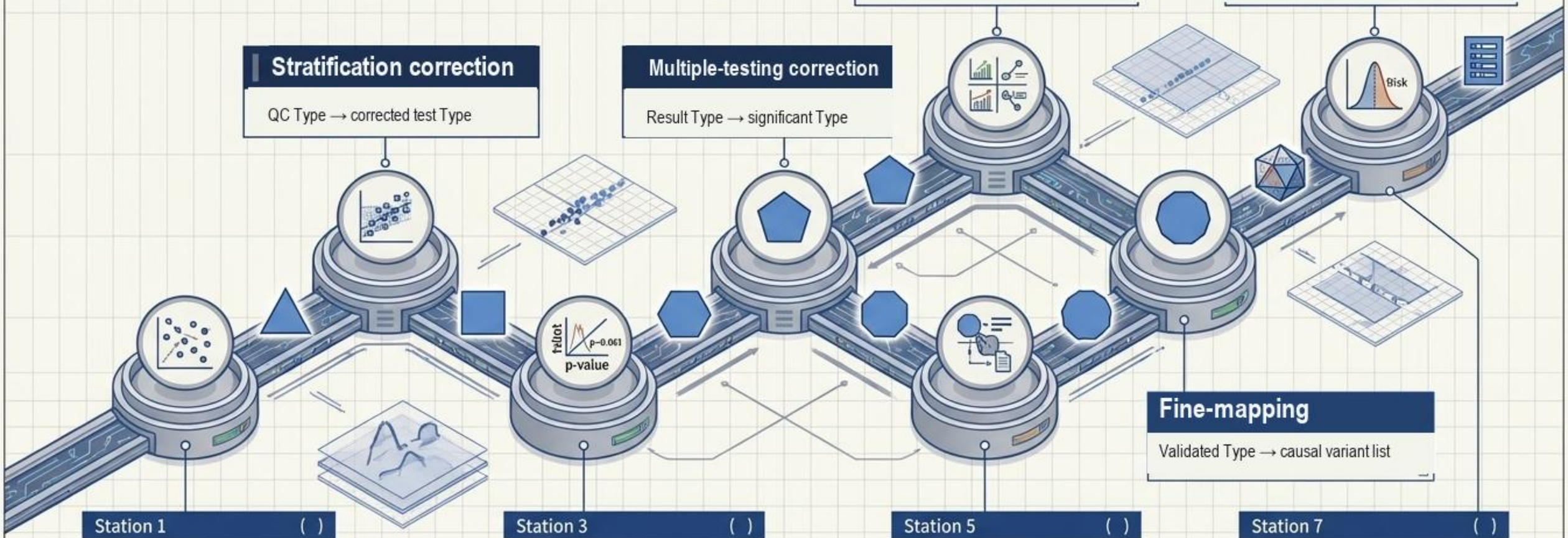
PCA & mixed models:
Subtract the container effect; align the containers.

Never compare numbers directly between containers that are not aligned.

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Time Type: the seven-step GWAS function chain

Function set: analysis · inference · cognition · design; no skipping.



Station 1 ( )

Quality control (QC)

Raw data Type → post-QC Type

Station 3 ( )

Association testing

Corrected Type → result Type

Station 5 ( )

Replication & meta-analysis

Significant Type → validated Type

Station 7 ( )

Polygenic risk score (PRS)

Variant list → risk-score Type

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Four questions per function: execution criteria at every step

UI Dashboard

Function card 1: Association testing

Name	→	Regression test (additive model)
Input Type	→	Genotype matrix (0/1/2) + phenotype + covariates (PC) 
Output Type	→	One row per locus: association statistics (β, p-value)
Criterion	→	Normal residuals, linearity, threshold (5×10^{-8})

UI Dashboard

Function card 2: Polygenic risk score (PRS)

Name	→	Effect-size-weighted composite Type
Input Type	→	GWAS effect sizes + target-cohort genotypes 
Output Type	→	Continuous risk score
Criterion	→	Cross-ancestry recalibration, R^2, AUC

Side Note

⚠ Criteria must be set before analysis. Setting the [ ] criterion afterwards is p-hacking.

MIND AI

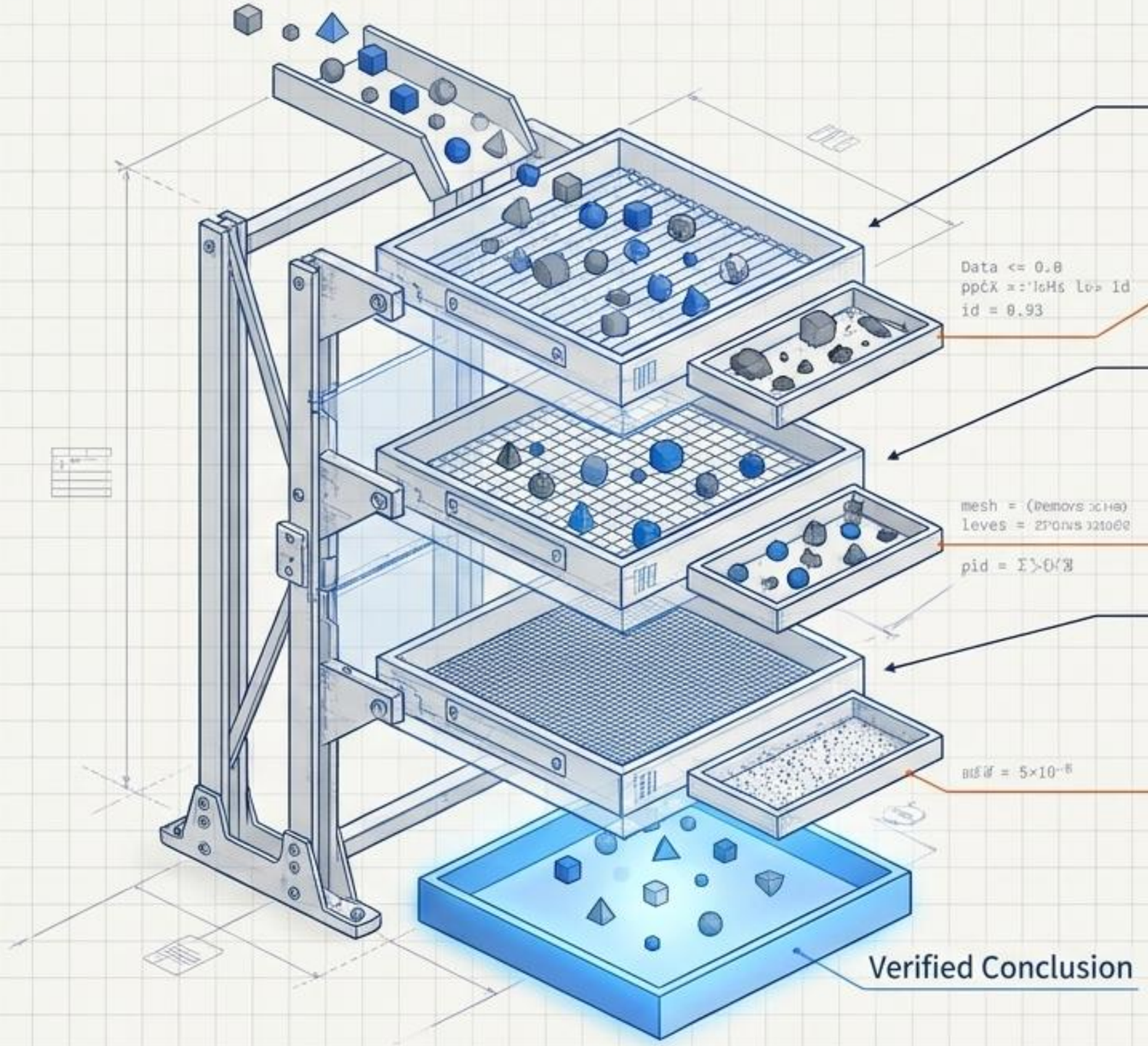
Four questions per function: no criterion, no sign-off

Association Test

Name	Association-test function
Input Type	Adjusted data Type (genotype + phenotype + first 10 PCs)
Output Type	Association result Type (one row per locus)
Criterion	$p < 5 \times 10^{-8}$

5×10^{-8} is not a magic number; it is the abstraction layer's strictest gate for one million independent effective tests. The criterion must match how many questions you actually asked.

Three-layer judgment: the filter before you hand over results



Container layer (Identity Check)

Which cohort id does the data come from?

Has population stratification (PCA) been cleaned?

⚠ Filters out: Population Stratification

Composition layer (Structure Check)

Are all essential parts present?

Has QC (Remove 3 levels) actually been executed?

⚠ Filters out: Poor Data Quality / Missing Covariates

Abstraction layer (Criteria Check)

Has it reached the 5×10^{-8} threshold?

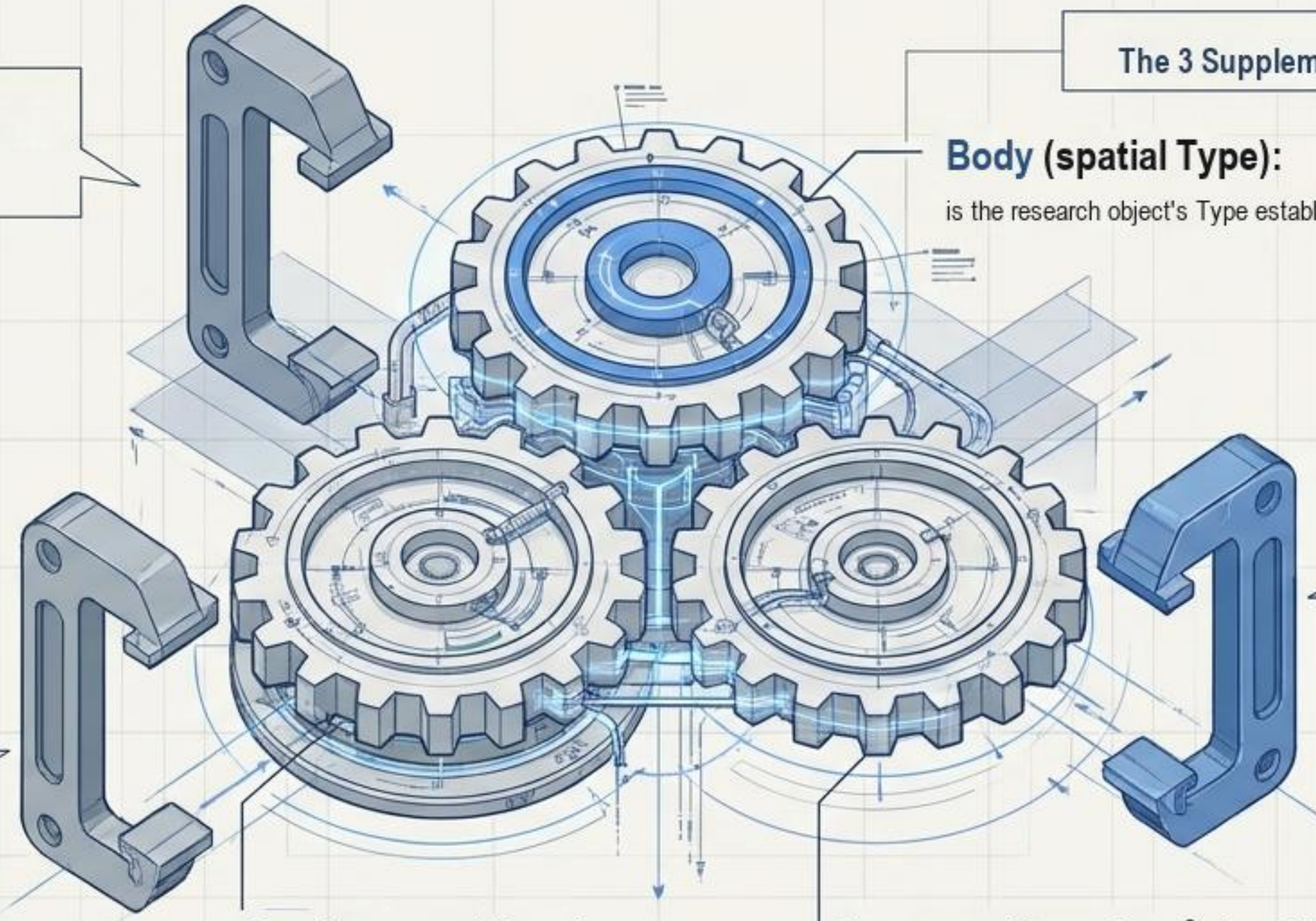
Is the direction consistent in an independent cohort?

⚠ Filters out: Statistical Flukes

Conclusion: all three layers together — none can be dropped. Concluding from a p-value alone is the most dangerous misuse.

MIND AI

The overall architecture of accurate use: Body — Use — Accuracy



The 3 Supplements (rule · structure · identity)

[ ] Rule : lock criteria
before results; no tampering.

Body (spatial Type):
is the research object's Type established?

{ } Structure : BOM
and a remove-fallback list, so no component is missed.

( ) Identity : lock the
data source id(C), and prevent sample overlap.

Use (temporal Type):
are the four questions written for every step?

Accuracy (three-layer check):
does it pass the review check?

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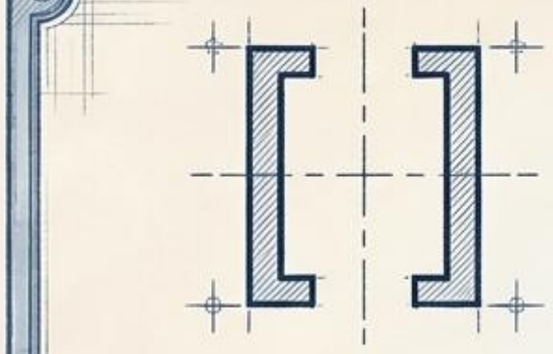
Diagnostic matrix: 16 common GWAS misuses and corrections

Type Errors	Vague phenotypes →	Fix: set the dimension scale first	Rules set after data →	Fix: register rules in advance
	Missing parts →	Fix: complete the BOM	Hard-coded classes →	Fix: open-ended rules
Number Errors	p as effect size →	Fix: report p and β separately	Copied / winner's curse	Fix: independent replication
	Numbers without form →	Fix: stratified test selection	MAF ignored →	Fix: label container and units
Container Errors	Residual stratification	Sample overlap	PRS across containers	Batch effects
	Fix: PCA + mixed model	Fix: cross-study id dedup	Fix: recalibrate	Fix: batch covariates
Reasoning Errors	Association as cause	One-layer judgment	Sloppy replication	Selective reporting
	Fix: Mendelian randomization	Fix: three-layer joint check	Fix: check direction afresh	Fix: publish all results

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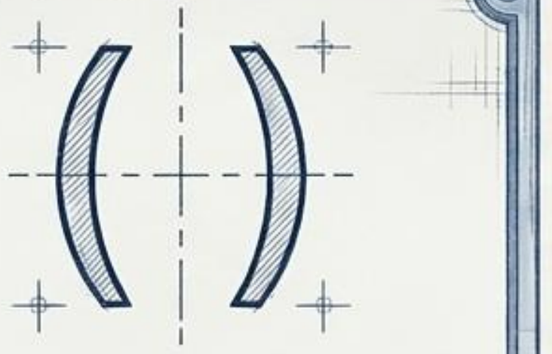
Eight checks before delivery: the acceptance line for the three supplements

Precision = verifiable precision. It passes only when someone else can tick every line and reproduce your reasoning.



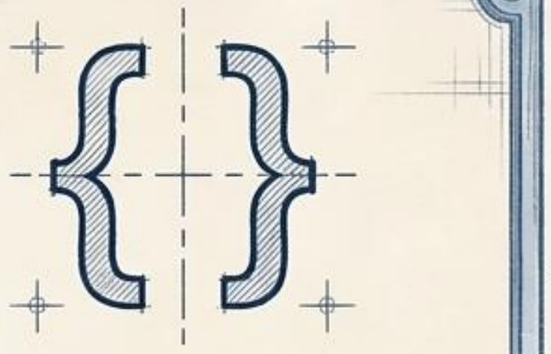
Rule supplement

- ☐ Is the phenotype Type (dimensions/scale) defined before analysis?
- ☐ Was the criterion (QC / significance threshold) written before seeing results?



Identity supplement

- ☐ Is the study object stated? (one sentence naming the whole being analysed)
- ☐ Is the container ID labelled? (sample overlap already checked)



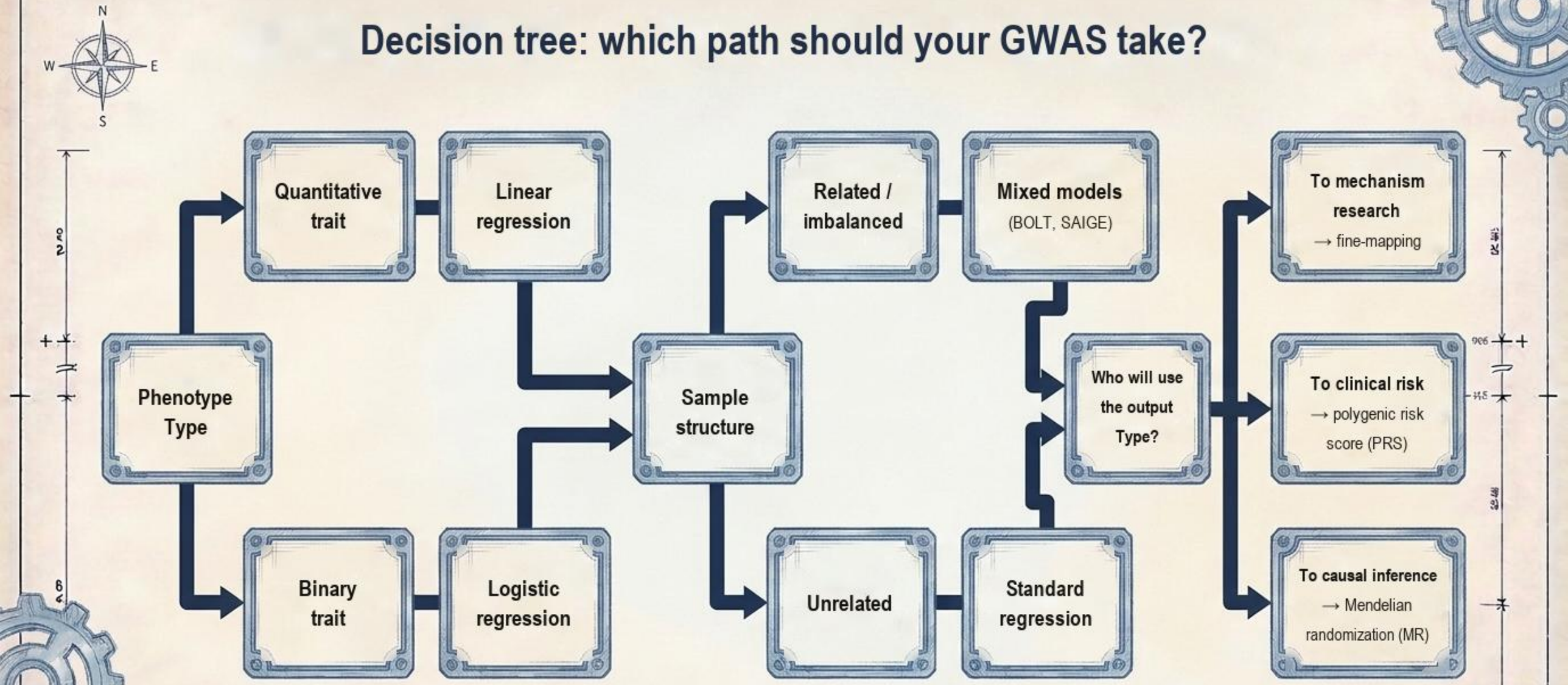
Structure supplement

- ☐ Are the remove levels applied? (QC exclusions logged item by item)
- ☐ Are the brackets used correctly? (parts, contents, criteria each in place)

↖ Rigor = 100%

Verified

Decision tree: which path should your GWAS take?



↖ Rigor = 100%

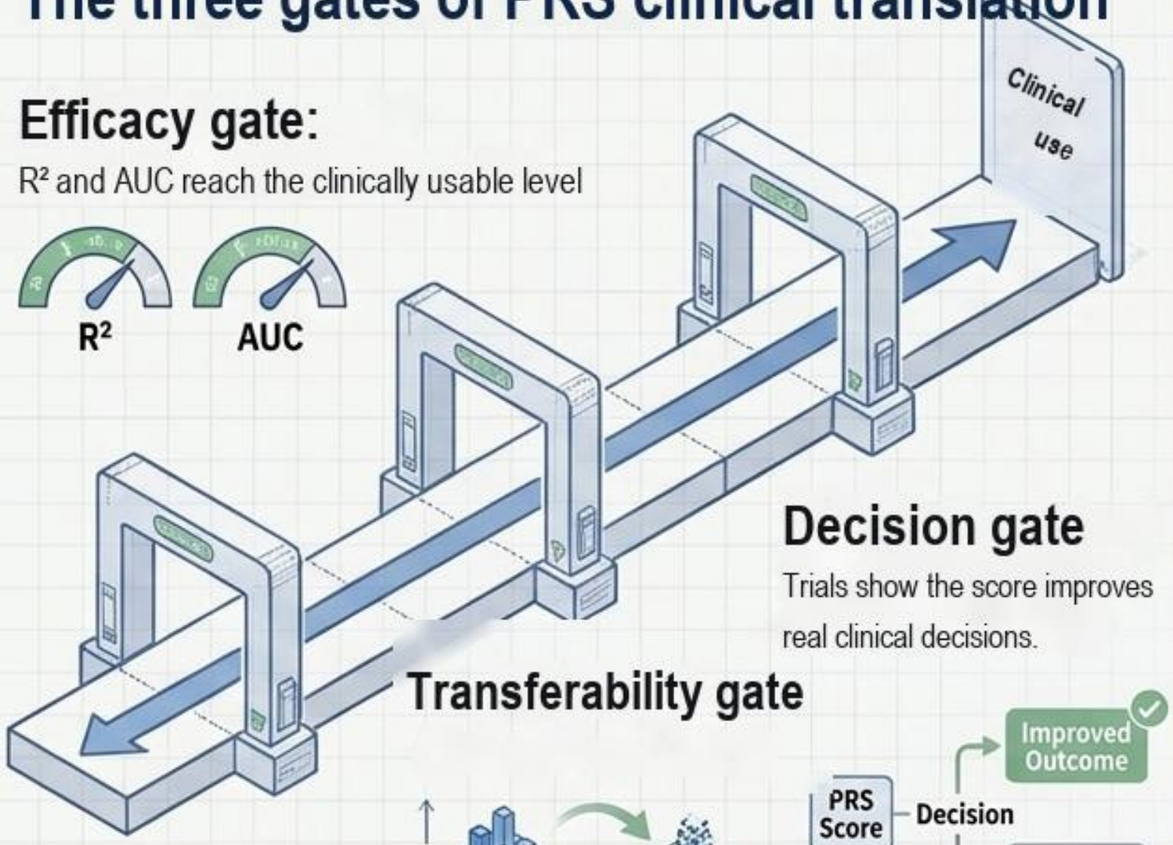
Verified

Clinical translation & container justice (ethics)

The three gates of PRS clinical translation

Efficacy gate:

R^2 and AUC reach the clinically usable level



Decision gate

Trials show the score improves real clinical decisions.

Transferability gate





Efficacy gate

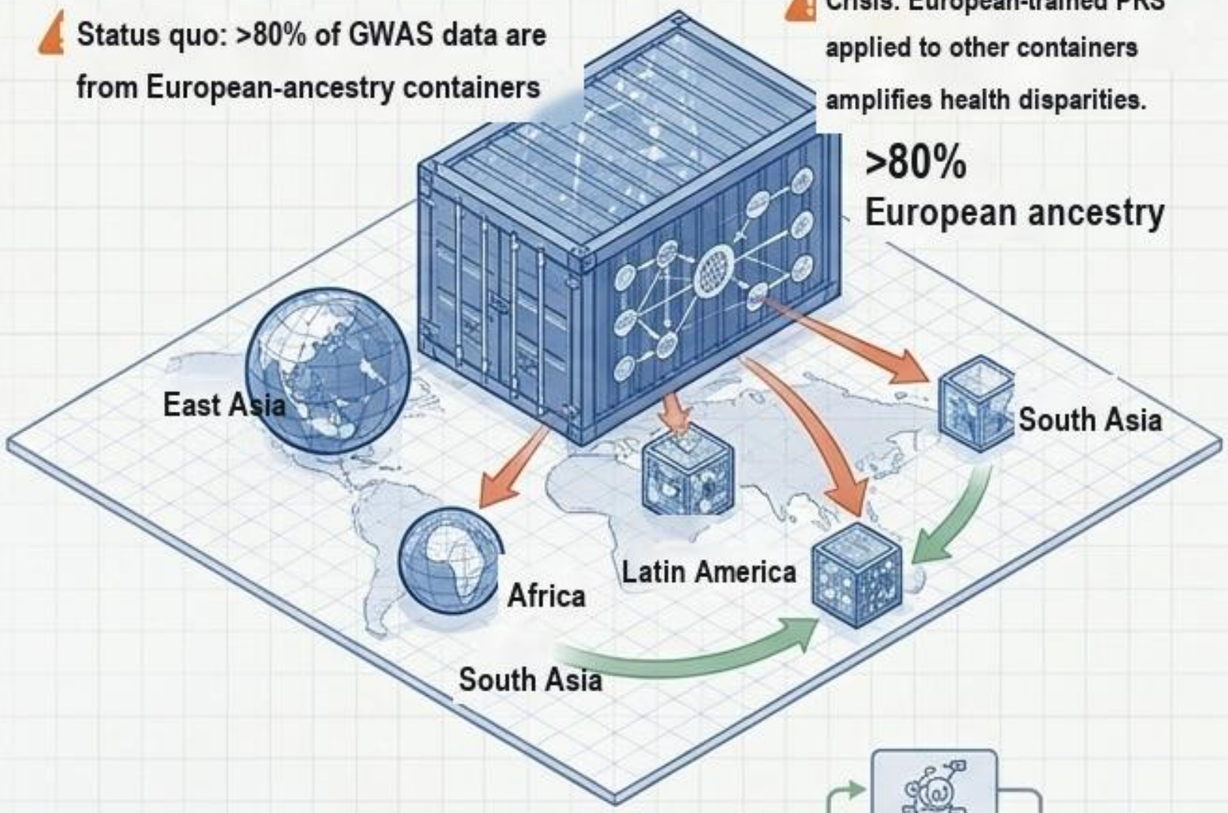
R^2 and AUC reach the clinically usable level

fixes cross-ancestry effect attenuation; re-estimated locally

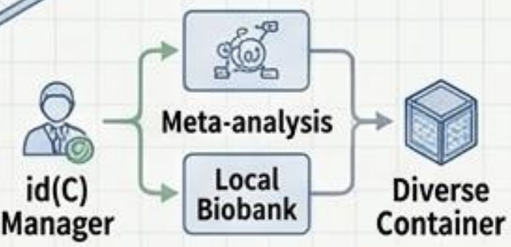
Multi-ancestry and container justice

Status quo: >80% of GWAS data are from European-ancestry containers

Crisis: European-trained PRS applied to other containers amplifies health disparities.



Formal solution: id(C) managers hold a fiduciary duty, driving local Biobank and pluralistic meta-analysis.



Acceptance criteria and the road to AGI unity

Evolution of a GWAS project

Inheritable:

Form and function archived in full, precisely reusable across teams.

Auditable:

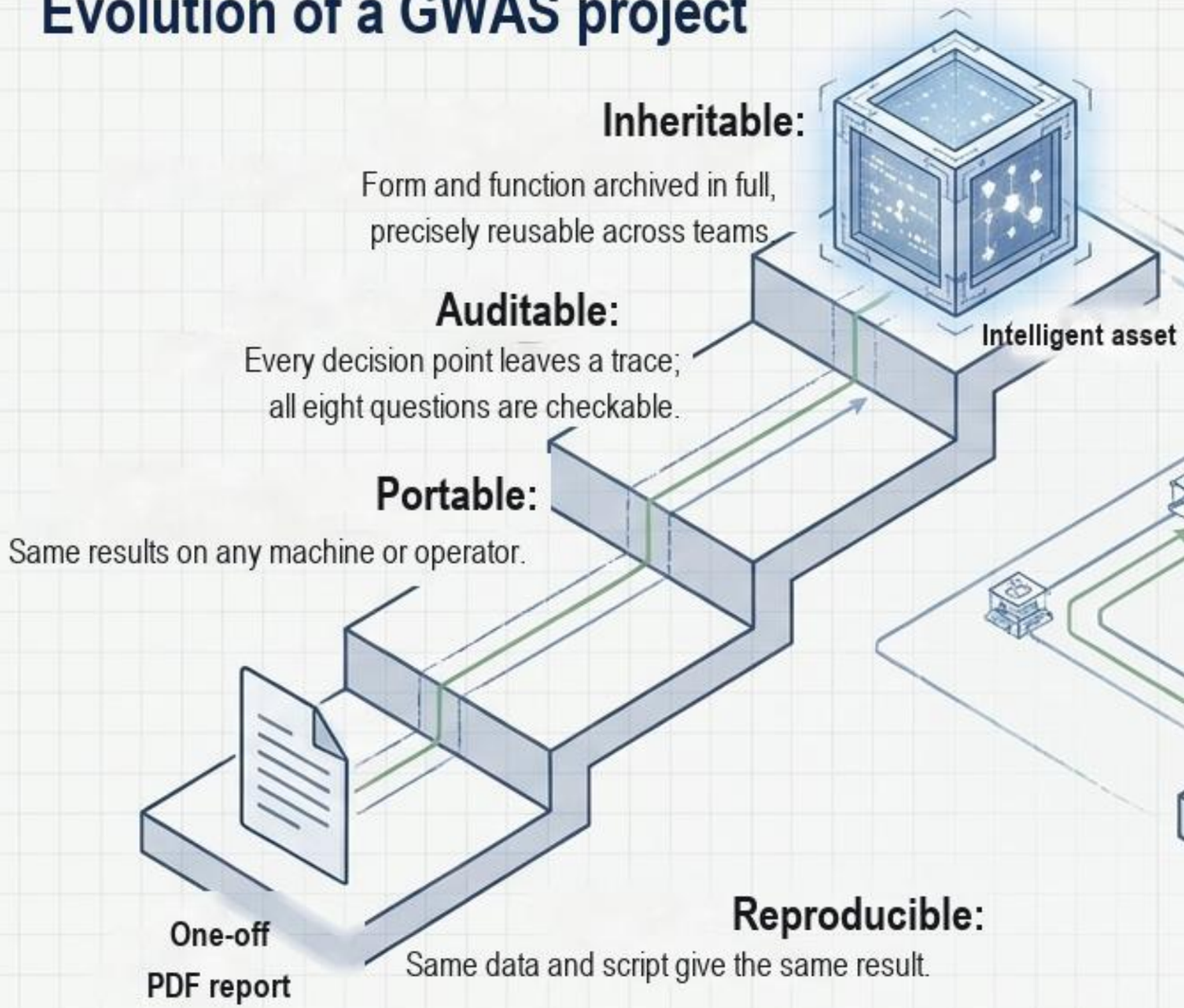
Every decision point leaves a trace; all eight questions are checkable.

Portable:

Same results on any machine or operator.

Reproducible:

Same data and script give the same result.



Evolution of an AI platform

Unified intelligence:

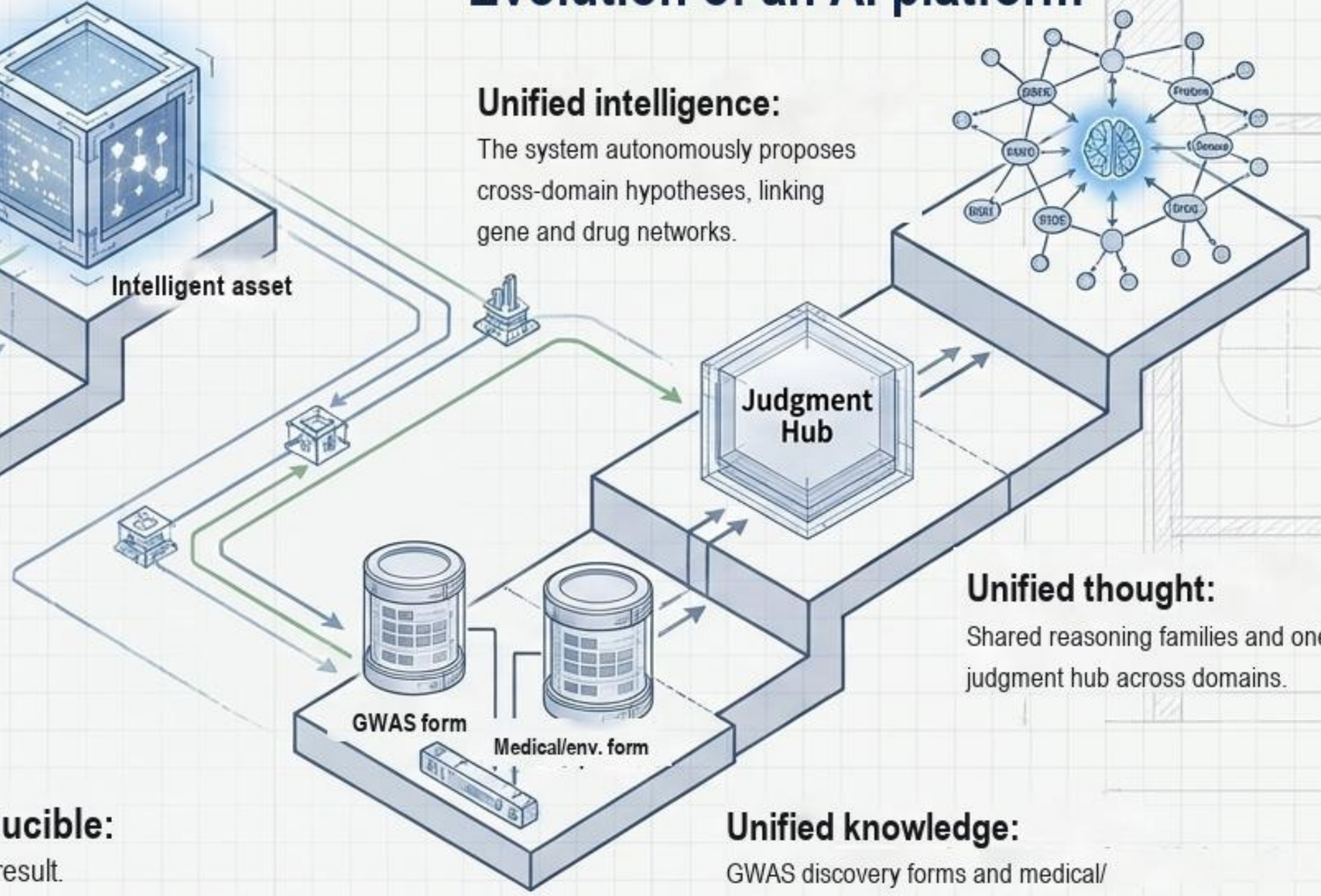
The system autonomously proposes cross-domain hypotheses, linking gene and drug networks.

Unified thought:

Shared reasoning families and one judgment hub across domains.

Unified knowledge:

GWAS discovery forms and medical/ environmental forms in one repository.



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Final acceptance: turn GWAS into an inheritable intelligent asset

Reproducible

- Same data, same script, same result.

Portable

- New machine or new operator, same result.

Auditable

- Criteria and decision points traced; the eight questions checkable.

Inheritable

- Type library and function cards complete; newcomers reuse correctly.



“Form first, then numbers. See GWAS clearly, and turn it into a teachable, trainable, verifiable intelligent asset.”

Rigor = 100%

Verified

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The pre-flight checklist before delivery

- [ ] **1. Is the study object stated?** (cohort + phenotype + generation explicit)
- [ ] **2. Is the phenotype Type specified?** (dimensions, scale, defined before analysis)
- [ ] **3. Are the brackets used correctly?** ({ } parts, ( ) contents, [ ] criteria each in place)
- [ ] **4. Is the container id labelled?** (prevent sample overlap)
- [ ] **5. Are the remove levels applied?** (QC exclusions logged by category)
- [ ] **6. Was the criterion fixed first?** (never change the threshold after seeing results)
- [ ] **7. Is every step testable?** (the script is reproducible)
- [ ] **8. Have all three layers been checked?** (container, composition, abstraction all pass)

Precision = verifiable precision

Building the library: an eight-week rollout and two-track path

Weeks 1–2



Establish Types: deliver phenotype-Type cards, the BOM, and the remove-level table.

Weeks 3–4



Run functions: deliver the first four function cards (QC, stratification correction, testing, ...).

Weeks 5–6



Acceptance and debugging: run all six steps end to end, clearing the 16-misuse matrix.

Weeks 7–8

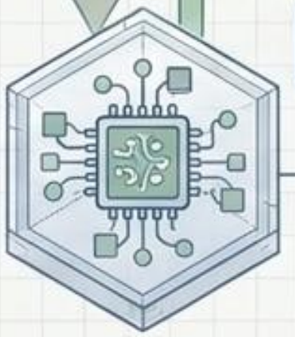


Build the library: complete one project's full report and deliver the first internal Type and function libraries.



Biomedical researchers

Focus: phenotype definition, criterion setting, clinical translation.

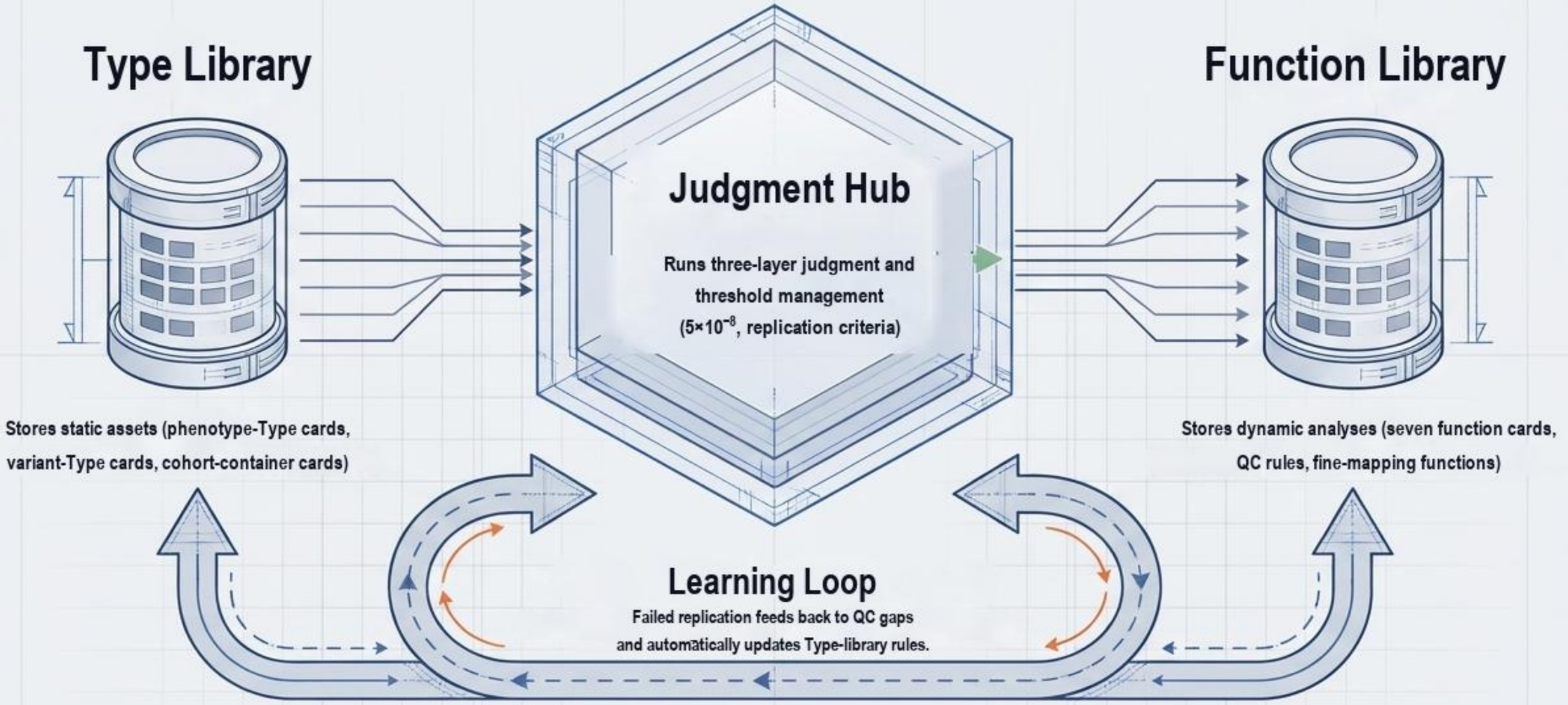


Data analysts

Focus: executing the function chain, reproducible scripts, QC standards.

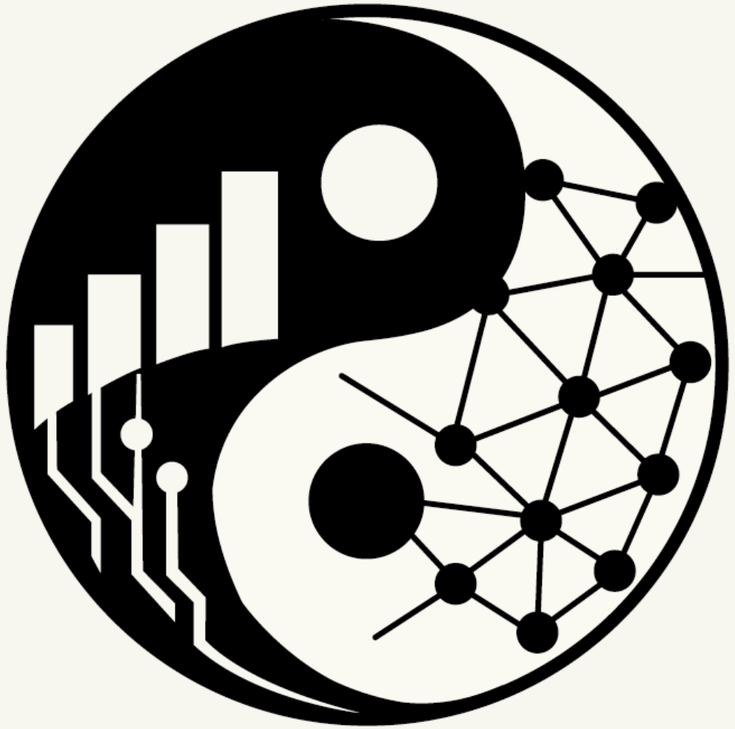
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System integration: the Mind AI infrastructure



Turn GWAS output from a one-off PDF report into an intelligent asset that machines can reason over and learn from.

LonGreat Vision: to create intelligent life that understands the essence of the world



MIND AI

"We aim to make artificial intelligence not merely a tool, but a system of intelligent life that truly possesses the abilities to understand, reason, learn, and create."

From the unification of knowledge, to the unification of thought, and finally toward the unification of intelligence – this is the eternal path to AGI that LonGreat pursues.

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